



Sherwin Williams Recommendation System: A Data-Driven Approach to Paint Selection

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Background



- ❑ Lead-based paint was widely used for decades in the U.S.
 - ❑ Rich color & durability
- ❑ 1970s public health experts began to realize it caused lead poisoning (very bad for children)
 - ❑ Neurological problems, stunted growth, learning disabilities, etc
 - ❑ 1978: Consumer Product Safety Commission banned lead-based paint



Methodology

Data Collection

PDS + Paint Grades
Guide
Extract + Merge
Product Attributes

Feature Engineering

Durability, coverage, VOC, Dry time
QSum (composite score)

Clustering

K-means Grouping
Reduced to 3 interpretable
clusters

Filtering & Ranking

User inputs (project location,
price, color)
Ranked by Qsum

Output (Flask App)

Personalized Paint
Recommendation

Product_ID	Product_Name	Interior	Exterior	VOC_Value	Recoat_Dry_Time	Coverage_Min	Coverage_Max	Cleanup	Price_Tier	Type
1	Promar 400	TRUE	FALSE	49	4	350	400	Water	Basic	Wall
2	Promar 200	TRUE	FALSE	49	4	350	400	Water	Basic	Wall
3	Promar Ceiling	TRUE	FALSE	49	4	350	400	Water	Basic	Ceiling
4	Extreme Cover	TRUE	FALSE	49	4	350	400	Water	Basic	Wall
5	Painters Edge	TRUE	FALSE	49	4	350	400	Water	Basic	Wall
6	Midsouth	TRUE	FALSE	80	4	325	400	Water	Basic	Wall
23	A100	FALSE	TRUE	49	4	350	400	Water	Basic	Wall
7	SnapDry	TRUE	TRUE	49	2	350	400	Water	Mid Range	Trim
8	Cashmere	TRUE	FALSE	49	4	350	400	Water	Mid Range	Wall

Product_ID	Appearance	Easy_Clean	Durability	Mold_&Mildew_Resistance	Coverage	Application	QSum
1	1	1	1	1	1	1	6
2	2	1	1	2	3	3	12
3	2	1	1	1	3	3	11
4	2	1	2	1	2	2	10
5	2	1	1	1	2	1	8
6	2	1	1	1	2	1	8
7	3	3	3	1	2	2	14
8	3	1	2	2	2	3	13
9	2	2	2	2	2	2	12
10	2	3	3	3	2	2	15

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6	Extra White	id of application and as that originally used
7	Extra White	id of application and as that originally used
7	Deep	tion was by spray, up paint by running it will help touch up
7	Ultra-deep	
8	Extra White	Williams.com

continued on back

Feature Engineering

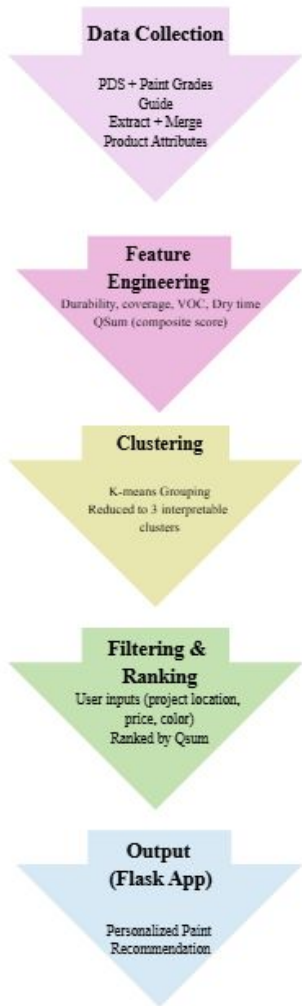
- ☆ = numeric value
- Qsum = summary of quality metrics
- Handle missing/inconsistent values
- Bases (Extra white, Deep, Ultra-deep, Accent) → light colors, dark colors, very dark colors, bright colors
- Interior = (True/False)
- Exterior = (True/False)

Clustering

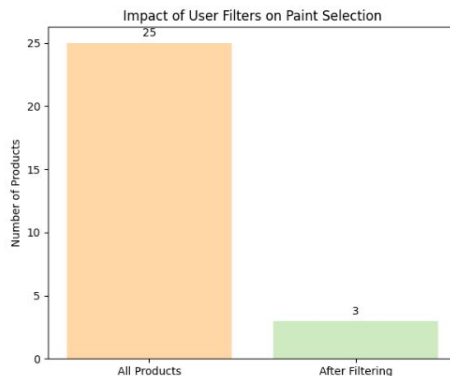
- Starting point of 4 clusters & tested
 - Overlap in clusters, 3 for more meaningful groups: Exterior/Heavy Duty, Premium/Low VOC, Budget/ Contractor Grade

Cluster	
0	2
1	11
2	8
3	4

Cluster	
0	12
1	11
2	2



Methodology



Filtering & Ranking

- ☐ Interior / Exterior
- ☐ Paint Type (Wall, Trim, Ceiling)
- ☐ Price Tier (\$, \$\$, \$\$\$)
- ☐ Color Intensity (bases filtering)
- ☐ Sheen
 - ☐ Qsum = asc. Values

Flask

- ☐ Tasked w/ taking user input
- ☐ Runs through filtering and ranking
- ☐ Clusters assigned to each paint (similarity comparisons)

Sherwin-Williams Paint Recommender

Choose the project needs below to find matching paint products.

Where will the paint be used?

Interior

Project surface type

Trim

Budget level

\$\$\$

What kind of color are you planning to use?

Light colors / whites

Finish / Sheen

Semigloss

Find Paints

RECOMMENDED PRODUCT

Proclassic Latex

★★★★☆
Quality Score: 10

Trim • \$\$\$

Available sheens: Satin, Semigloss, Gloss, High Gloss

ProClassic Waterborne Interior Acrylic Enamel is a high-quality coating that provides a durable, non-yellowing finish with excellent flow and leveling for a smooth, rich result comparable to an alkyd enamel, ideal for trim, molding, cabinets, doors, windows, kitchens, bathrooms, laundry rooms, locker rooms, and high-traffic areas, and suitable for surfaces such as drywall, plaster, wood, metal, concrete, and masonry.

[Read less](#)



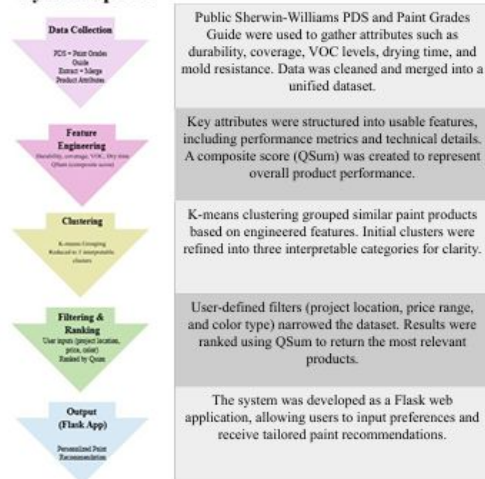
Interior Trim \$\$\$ Premium / Low VOC

Background & Overview

- Builds a paint recommendation system for Sherwin-Williams products using structured data, feature engineering, and clustering.
- Combines multiple datasets with user-driven filters to deliver tailored recommendations through a Flask web app.
- After the 1978 ban on lead-based paint by the Consumer Product Safety Commission, product options expanded significantly (mold & mildew resistance, low-VOC, antimicrobial coatings, etc).
- This growth in industry created a complex decision-making environment, which this system simplifies into clear, user-friendly recommendations.

Methodology

System Pipeline



Results



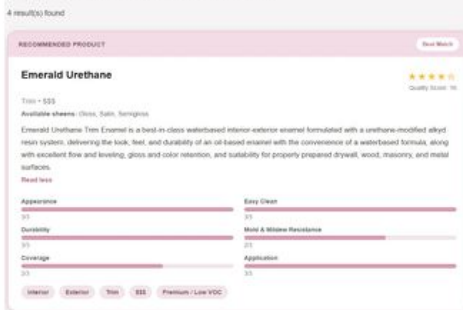
Figure 1: Heatmap of average engineered features across K-means clusters. Reveals distinct product groupings based on performance characteristics.

- Exterior / Heavy Duty**
Products with stronger exterior use characteristics
- Budget / Contractor Grade**
Lower overall performance scores and more basic options
- Premium / Low VOC**
Higher-performing products with lower VOC levels
- Premium / High Durability**
Strong overall performance with emphasis on durability

Figure 2: Product count before and after filtering. Demonstrates how user inputs reduce the search space and improve recommendation focus.

Figure 3: Example output from the Flask application showing recommended products ranked by performance (QSum) based on user-defined filters.

Recommended Paints



The results highlight the challenge of distinguishing signal from noise in product data. Although paint products often appear similar, the use of structured features and clustering reveals meaningful differences in performance and intended use. When combined with user-driven filtering, the system provides a clearer and more efficient way to navigate a complex product space.

Discussion & Conclusion

The system generates personalized paint recommendations based on user input, returning products that match selected criteria such as use case, price range, and color type. Results are ranked using a performance based score (QSum), ensuring higher-quality products appear first. Clustering further enhances the output by grouping similar paints, helping users compare options within meaningful categories and better understand trade-offs between products.

Implications:

- Transforms complex manufacturer data into user-friendly decision support.
- Reduces reliance on heavy product descriptions and technical documentation.
- Demonstrates how combining structured data, filtering, & modeling improves recommendation quality.

Limitations

- Clustering often reflects existing product similarities, offering limited additional differentiation.
- Recommendations depend on available data and may not capture all real-world factors (user preference, brand familiarity).
- The dataset is static and does not automatically update with newly released Sherwin-Williams products (recently introduced products like "Forte" and "Symmetry").

Future Directions

- Incorporate user preference weighting (prioritize durability vs cost).
- Expand dataset to include more products and real user feedback.
- Explore alternative models beyond K-means for more advanced product classification

